P.7 Inverse Trigonometric Functions

Name _____ Period ____ Date _____

AP Calculus P.7 · Student sheet

I can…

- find exact values of inverse trigonometric functions.
- evaluate composites of trigonometric and inverse trigonometric functions.
- solve trigonometric equations on $[0, 2\pi)$.

Vocabulary · fill in the missing word

Inverse sine function — $y = \sin^{-1} x$ _____ $x = \sin y$, where $-1 \le x \le 1$ and $-\pi/2 \le y \le \pi/2$

Inverse cosine function — $y = \cos^{-1} x$ _____ $x = \cos y$, where $-1 \le x \le 1$ and $0 \le y \le \pi$

Inverse tangent function — $y = \tan^{-1} x$ _____ $x = \tan y$, where $-\pi/2 < y < \pi/2$

Part 1 · Which angle?

Find every θ in $[0, 2\pi)$.

Equation	Angles from the unit circle	Calculator (radian mode)
$\sin\theta = 1/2$		$\sin^{-1}(0.5) =$
$\cos\theta = -1/2$		$\cos^{-1}(-0.5) =$
$\tan\theta = 1$		$\tan^{-1}(1) =$

Part 2 · Notes and examples (fill in each note, then work the example)

Inverse sine: $y = \sin^{-1} x$ means $x = \sin y$, where $-1 \le x \le 1$ and _____ $\le y \le \pi/2$.

Example 1

Find the exact value.

a. $\sin^{-1}\left(\frac{\sqrt{2}}{2}\right)$

b. $\sin^{-1}\left(-\frac{\sqrt{3}}{2}\right)$

c. $\sin^{-1} 0$

Ranges: $\cos^{-1} x$ is in $[0, \pi]$; $\tan^{-1} x$ is in the open interval (_____).

Example 2

Find the exact value.

a. $\cos^{-1}\left(-\frac{1}{2}\right)$

b. $\tan^{-1}(-1)$

c. $\tan^{-1}\sqrt{3}$

Inverse properties: $\sin(\sin^{-1} x) = x$ for $-1 \le x \le 1$, but $\sin^{-1}(\sin x) = x$ only for x in _____.

Example 3

Find the exact value.

a. $\sin^{-1}(\sin(\frac{9\pi}{8}))$

b. $\tan^{-1}(\tan(\frac{4\pi}{5}))$

Example 4

Find the exact value.

a. $\sin(\tan^{-1} 2)$

b. $\cos(\tan^{-1} 3)$

Solving trig equations: isolate the trig function (factor if needed), then find every solution in one _____.

Example 5

Solve on $0 \le \theta < 2\pi$.

a. $2\sin^2\theta - 5\sin\theta + 3 = 0$

b. $2\cos^2\theta - 7\cos\theta - 4 = 0$

Part 3 · Partner practice

Partner 1

Find $\cos(\sin^{-1}(\frac{1}{4}))$.

Partner 2

Solve $\tan\theta = -\frac{\sqrt{3}}{3}$ on $0 \le \theta < 2\pi$.